

Let $\delta(z_0) = \frac{e^{-E(z_0|c)}}{\mathbb{E}_{q(z_0|z_\ell)}[e^{-E(z_0|c)}]} - E(z_0|c)$ represent the difference between the ideal exponential weighting (normalized) and the approximate linear weighting. The term in the parenthesis can be viewed as $\mathbb{E}_{q(z_0|z_\ell)}[\delta(z_0)\epsilon]$. In high-dimensional latent spaces (i.e., when d is large), the noise vector ϵ typically has a magnitude of approximately $\sqrt{d}$ (i.e., $\|\epsilon\|_2 \approx \sqrt{d}$). Furthermore, the components of ϵ are largely independent. Due to the Central Limit Theorem and concentration inequalities, even small biases in the weighting function $\delta(z_0)$ can lead to a significant accumulated error when multiplied by a high-dimensional random vector. Specifically, if $\mathbb{E}_{q(z_0|z_\ell)}[\delta(z_0)] \neq 0$, the term $\|\mathbb{E}_{q(z_0|z_\ell)}[\delta(z_0)\epsilon]\|_2$ will tend to scale with $\sqrt{d}$ in expectation, as the contributions from different dimensions accumulate.

Therefore, there exists a constant $c > 0$ (which depends on the magnitude of the mismatch $\delta(z_0)$ and the properties of the noise distribution) such that:

$$\Delta_{\text{EBM}}(z_\ell) \geq \frac{c}{\sqrt{1 - \bar{\alpha}_\ell}} \sqrt{d}$$

This lower bound shows that the inaccuracies in energy guidance are exacerbated in high-dimensional latent spaces and as the diffusion process approaches z_0 (i.e., as $\ell \rightarrow 0$, $1 - \bar{\alpha}_\ell \rightarrow 0$, making the term $\frac{1}{\sqrt{1 - \bar{\alpha}_\ell}}$ large). This leads sampled trajectories to deviate significantly from the true data manifold. $\square$

A.2. Proof for Proposition on Manifold-Aware Guided Planning

Statement. Given a base diffusion planner ϵ_θ trained for classifier-free guidance, a learned energy function $E_\phi(z|c)$, and a manifold projection operator $\mathcal{P}_\mathcal{M}$, the manifold-aware guided planner, which combines EBM guidance with manifold projection, corresponds to sampling from a posterior distribution $p(z|y=1, z \in \mathcal{M}, c)$ that maximizes the likelihood of generating a successful and feasible goal-conditioned plan.

Proof. The manifold-aware guidance process can be viewed as implementing constrained Bayesian inference. We seek to sample from the posterior:

$$p(z|y=1, z \in \mathcal{M}, c) \propto p(y=1|z, c)p(z|c)\mathbf{1}[z \in \mathcal{M}]$$

where $\mathbf{1}[z \in \mathcal{M}]$ is the indicator function for the feasible manifold.

The two-step process approximates this constrained posterior:

1. The guided sampling step samples from $p(y=1|z, c)p(z|c)$, implementing the unconstrained Bayesian posterior. This is achieved by combining classifier-free guidance for the conditional term $p(z|c)$ and EBM guidance for the success term $p(y=1|z, c)$, as detailed in Proof A.3
2. The projection step enforces the manifold constraint $z \in \mathcal{M}$ by mapping to the closest point on the approximated manifold.

By the principle of alternating projections and the contraction property of projection operators, this two-step process converges to a point that balances optimality (high success probability) with feasibility (remaining on the manifold). The approximation error depends on the quality of the local manifold approximation, which improves with the number of neighbors k and the intrinsic dimensionality of the subgoal space. This shows that our combined guidance approach is a principled implementation of Bayes-optimal sampling, steering the generative process towards plans that are both relevant to the goal and likely to succeed, while remaining on the feasible manifold. $\square$

A.3. Proof for EBM-based Guidance

Statement. Given a base diffusion planner ϵ_θ trained for classifier-free guidance and a learned energy function $E_\phi(z|c)$ that estimates the probability of success conditioned on context c , the EBM-guided component of the planner corresponds to sampling from a posterior distribution that maximizes the likelihood of generating a successful, goal-conditioned plan.

Proof. The proof proceeds in two steps. First, we derive the theoretically optimal score function for sampling successful, goal-conditioned plans using Bayes' rule. Second, we show how our combined guidance mechanism implements an effective approximation of this optimal score.